INCH-POUND

MIL-L-757A
INT. AMENDMENT 2 (AR)
18 May 90

MILITARY SPECIFICATION

LEAD STYPHNATE, NORMAL

This Interim Amendment is approved for use within the U.S. Army Armament, Munitions and Chemical Command, with Military Specification MIL-L-757A dated 1 September 1967.

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* 2.2 Under this paragraph, delete: "49 CFR-71-90-Rules and Regulation...and other Dangerous Articles." and substitute the following:

"49 CFR 100-199 - Department of Transportation Rules and
(Para 173.74) Regulations for the Transportation of Explosives and other Dangerous Articles".

3.1 Appearance. Delete in its entirety and substitute the following:

"3.1 Appearance. The material shall be free from visible impurities and free from aggregates which are larger than the maximum allowable crystal dimensions permitted under para. 3.3."

3.2 Crystal form. Delete in its entirety and substitute the following:

"3.2 Crystal form. The material shall be crystalline and the longest dimension of any crystal shall not be greater than three times the next longest axis. Hexagonal plates are acceptable."

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* 4.3 Sampling. Delete in its entirety and substitute the following:

"4.3 Sampling. Obtain two randomly selected 30 gm samples from each batch of lead styphnate being submitted for inclusion in a lot. The samples shall be obtained from process vessel prior to packout using procedures decribed in ASTM E 300 for slurries.

* 4.3.1 Testing for batch acceptance. Delete in its entirety and substitute the following:

"4.3.1 Testing for batch acceptance. One sample obtained according to 4.3 from each batch shall be tested in accordance with applicable test methods for conformance with the requirements of color and appearance (see 3.1), crystal form (see 3.2), crystal dimensions (see 3.3), apparent density (see 3.5), and acidity (see 3.5). If any sample fails to meet any of these requirements, the batch represented by the sample shall be rejected. A rejected batch may be reworked to meet these requirements.

* 4.3.2 Testing for lot acceptance. Delete in its entirety and substitute the following:

"4.3.2 Testing for lot acceptance. The second samples from the batches not rejected under provisions 4.3.1 shall be placed in a suitable container of sufficient volume to hold a composite sample from all batches in a lot. The composite sample shall be vigorously stirred to assure homogeneity and two subsamples of approximately 30 gm each shall be obtained from the composite samples. Both subsamples shall be tested in accordance with applicable test methods of 4.5 to determine conformance with the requirements of granulation (see 3.4), material insoluble in ammonium acetate (see 3.5), nitrogen content (see 3.5), and instantaneous flash point (see 3.5). If either subsample fails to meet any of the test requirements, the lot represented by the composite sample shall be rejected."

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* 4.5.2 Appearance. Delete in its entirety and substitute the following:

"4.5.2 Color and appearance. Take a portion of the well stirred wet sample from 4.3 and spread over an area of approximately 4 sq. cm. on a glass microscope slide. Allow the sample to air dry. Examine the material microscopically for color and impurities under daylight illumination."

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* 4.5.3 Crystal form. Delete in its entirety and substitute the following:

"4.5.3 Crystal form. Using the slide prepared in 4.5.2, observe the crystalline form (shape) under a microscope using a magnification of about 100X under reflected light, and optionally, take a photomicrograph at about 50X magnification."

* 4.5.4 Crystal dimensions. Delete in its entirety and substitute the following:

"4.5.4 Crystal dimensions. Using the slide prepared in 4.5.2, measure the longest dimension of the larger crystals observed when scanning the field with the aid of a mechanical stage. Use transmitted and reflected light and an ocular micrometer. Consider the longest dimension to be the distance between the most remote points of any one crystal."

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* Add new paragraph 4.5.6.1:

"4.5.6.1 Apparent density (Alternate Method). Transfer approximately 2 to 3 grams of the wet sample obtained in 4.3 to a Buchner funnel fitted with medium porosity filter paper. Rinse the wet sample with 2B alcohol and displace water. Rinse the sample with n-butyl alcohol to displace the 2B alcohol. Transfer the sample to a 5 or 10 mL graduate, having 0.1 mL graduations. Rinse down the sides of the cylinder with enough n-butyl alcohol to slurry the sample. Stir with a wooden stick, or agitate by some means in order to free any trapped air bubbles. Fill to the top cylinder graduation with n-butyl alcohol as accurately as possible, being sure the sides are rinsed clean. Weigh the cylinder and contents to the nearest milligram (0.001 gm). Record this weight as W1. Allow the cylinder contents to settle for a minimum of 2 hours, then note volume occupied by the sample. Record this volume as V. Determine the weight of the cylinger filled with n-butyl alcohol only (no sample). Record this weight as W2.

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Calculate the apparent density as follows:

Apparent density = $R(w1 - W2)/V$

$R = Ds/(Ds - Db)$

Ds = Density of normal lead styphnate

Db = Density of n-butyl alcohol"

* Add new paragraph 4.5.7.1:

"4.5.7.1 Acidity (Alternate Method). Transfer approximately 50 mL of the water covering the sample obtained in 4.3 to a 200 mL beaker. Determine the pH value of this solution in accordance with Method given in ASTM - E70."

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* Add new paragraph 4.5.11.4:

"4.5.11.4 Nitrogen (Alternate Method). Determine the nitrogen content of the lead styphnate in accordance with Method 409.1 of MIL-STD-650."

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* Paragraphs 5 and 5.1: Delete in their entirety and substitute new paragraphs 5, 5.1, 5.1.1, 5.1.2 and 5.1.3.

* "5. PACKAGING

* 5.1 Packing. Normal lead styphnate shall be packed in a wet condition as directed in 49 CFR 100-199, para 173.74. A mixture of water plus denatured alcohol in such proportions to prevent freezing shall be always used unless otherwise directed.

* 5.1.1 Level A. A maximum of 25 lbs (dry weight) normal lead styphnate shall be packed wet (per para 5.1) directly into a cambric diaper or similar cloth bag. A cap of the same diameter and fabric as the bag shall be placed inside the bag and over the normal lead styphnate. The bag shall

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then be tied closed and placed inside a bag made of conductive velostat. This velostat bag shall then be filled with the water/alcohol mixture which meets requirements of para 5.1. This water/alcohol mixture must completely cover the bag contents. The velostat bag is then tied closed and placed inside a 4 oz. or heavier duck bag. A cap of the same material and of the same diameter as a duck bag shall be placed over the velostat bag. The duck bag is then securely tied closed. This bag is now the unit bag which shall be marked in accordance with para 5.2. An appropriate number of unit bags whose combined total dry weight of normal lead styphnate shall not exceed 150 lbs is placed into a rubber cloth bag. Inside this bag and over the unit bags of normal lead styphnate shall be placed a cap of the same fabric and same diameter as this bag. This rubber cloth bag shall be filled with a water/alcohol mixture meeting requirements of para 5.1 so that the unit bags are completely covered. The bag is then securely tied closed. This rubber cloth bag is placed inside a strong grain bag which must then be securely tied closed. The grain bag and its content must be packed in the center of a Department of Transportation Specification 5 or 5B metal drum or barrel which has been lined with a heavy, close-fitting, jute bag. The grain bag shall be entirely surrounded by not less than three inches of well-packed saw dust saturated with the previously referenced water/alcohol mixture. The jute bag shall be closed by secure sewing toprevent the escape of the sawdust. After closure of the barrel or drum, the package will be inspected carefully and all leaks stopped. The dry weight of normal lead styphnate in one outside container shall not exceed 150 pounds.

* 5.1.2 Level B. Level B shall be the same as Level A.

* 5.1.3 Level C. Level C shall be the same as Leval A except DOT specification 17H metal drum (single trip), as specified in CFR, Title 49, may be used."

* 5.2 Marking. Delete in its entirety and substitute the following:

"5.2 Marking. Markings shall be in accordance with CPR 49 and MIL-STD-129. Markings shall not be less then one-fourth (1/4) inch, nor more than one-half (1/2) inch in height. DOT markings must be one-half (1/2) inch in height. Ink shall conform to that specified in MIL-STD-129 and a contrasting color to the bag.

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* 5.2.1 Unit container. Each unit bag as defined in 5.1.1 shall be marked with the following:

a. Lead Styphnate, Normal
b. Quantity (dry weight) in pounds
c. Lot number

As an alternative, each unit bag may be identified by using a waterproof tag that is marked with the above information, imprinted with a suitable water/alcohol proof ink.

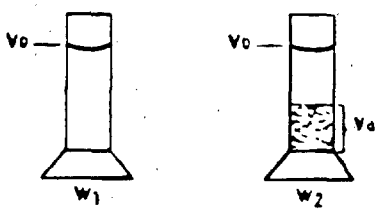
* 5.2.2 Exterior container. The side and top of each drum or barrel shall be marked: "INITIATING EXPLOSIVE - DANGEROUS - DO NOT STORE OR LOAD WITH ANY HIGH EXPLOSIVE".

The following additional information will be marked on the side of each drum or barrel:

a. NSN/NATO Stock number, if available
b. Lead Styphnate, Normal
c. Quantity (net weight in pounds)
d. Lot number
e. Gross weight
f. Any special cautionary marking and labeling"

DERIVATION OF APPARENT (BULK) DENSITY FORMULA

W_1 = weight of V_o mL liquid + G



G = weight of graduate

$W_1 = D_L V_o + G$

where:

D_L = density of liquid

W_2 = weight of sample + (weight of V_o mL liquid minus weight of liquid displaced by V_s mL of powdered sample) + G

$= W_s + D_L V_o - D\ V + G$

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$$W_2 = W_s + D_L(V_o - V_d) + G$$

but:

$W_s = D_s V_s$ = weight of powdered sample

where:

D_s = density of the powdered material

V_s = true volume of powdered material

V_d = volume of liquid displaced by the powdered material

but:

$$V_s = V_d$$

substitute into Eqn (2):

$$W_2 = D_s V_s + D_L(V_o - V_s) + G$$

subtract Eqn (1) from Eqn (4)

$$(W_2 - W_1) = D_s V_s + D_L(V_o - V_s) + G - D_L V_o - G$$

$$= D_s V_s - D_L V_s$$

$$= V_s(D_s - D_L)$$

$$(W_2 - W_1)/(D_s - D_L) = V_s$$

multiply Eqn (8) by D_s and rearrange

$$D_s V_s = D_s (W_2 - W_1)/(D_s - D_L)$$

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Divide Eqn (9) by Va:

$$D_s V_s/V_a = [D_s(W_2 - W_1)]/[(D_s - D_L)V_a]$$

where:

V_a = apparent volume of powdered material

but,

$D_s V_s/V_a$ = Apparent density of powdered material

$$= [D_s/(D_s - D_L)][(W_2 - W_1)/V_a]$$

NOTE: The labels, W_1 and W_2, have been interchanged from those in paragraph 4.5.6.1.

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